$||f(x) - f(y)|| < c||x - y||;\ \ c \in (0,1)$
$f : X \to X, X$ spatiul liniar; f se numeste contractie

Daca $f : X \to X$ este contractie $\Rightarrow \exists!\ c\ a.\ i.\ f(c) = c$
$x_{n+1} = f(x_n) \leftarrow$ **metoda aproximatiilor succesive**

$f : K^n \to K^n\ \ f(x)\ = \mathrm{Tx} + c$

$\mathrm{Ax} = b \Rightarrow x = \mathrm{Tx} + c$
$||f(x) - f(y)|| = ||\mathrm{Tx} + c - \mathrm{Ty} - c|| = ||T(x + y)|| \leq ||T|| * ||x - y|| \Rightarrow ||T|| \leq 1 \Leftrightarrow f$ contractie
$x^{(0)}$
$x^{(k+1)} = f(x^{(k)}) = \mathrm{Tx}^{(k)} + c$
x^* solutie $\Rightarrow x^* = \mathrm{Tx}^* + c$
$||x^* - x^{(n)}|| \leq \dfrac{||T||^n}{1 - ||T||} ||x^{(1)} - x^{(0)}||$

Metoda aproximatiilor succesive

$x^{(0)}$
$x^{(k+1)} = \mathrm{Tx}^{(k)} + c$
(1) converge $\Rightarrow \rho(T) < 1$

Ax = b

A = M - N

M inversabila, usor de inversat

$(M - N)x = b \Leftrightarrow \mathrm{Mx} - \mathrm{Nx} = b \Leftrightarrow \mathrm{Mx} = \mathrm{Nx} + b \Leftrightarrow x = \underset{T}{M^{-1}\mathrm{Nx}} + \underset{c}{M^{-1}b}$

$x = M^{-1}(\mathrm{Nx} + b) = M\backslash(N * x + b)$
$||x^{(k)} - x^{(k-1)}|| < \dfrac{1 - ||T||}{||T||} \varepsilon_a\ + ||x^{(k)}|| \varepsilon_r$
$\varepsilon_a -$ eroare absoluta, $\varepsilon_r -$ eroare relativa

$b - \mathrm{Ax}^{(k)} = r^{(k)}$
$||b - \mathrm{Ax}^{(k)}|| \to$ eroare absoluta
$\dfrac{||b - \mathrm{Ax}^{(k)}||}{||b||} < \varepsilon_r$

$$a_{i1}x_1 + a_{i2}x_2 + \ldots + a_{ii}x_i + \ldots + a_{in}x_n = b_i$$

$$x_i^{(k+1)} = \frac{1}{a_{ii}}\left(b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k)} - \sum_{j=i+1}^{n} a_{ij}x_j^{(k)}\right)$$

$$x_i^{(k+1)} = \frac{1}{a_{ii}}\left(b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij}x_j^{(k)}\right)$$

```
function [x,ni] = Jacobi(A,b, x0, err, it_max)
    if nargin < 5, it_max = 50; end
    if nargin < 4, err=1e-3; end
    if nargin < 3, x0=zeros(size(b)); end
    [m,n] = size(A);
    M=diag(diag(A));
    N=M-A;
    x=x0(:);
    for i = 1:it_max
        x0=x;
        x = M\(N*x0+b);
        if norm(x-x0,inf) < err*norm(x,inf)
            ni=i;
            return
        end
    end
    error('iteration number exceeded');
end
```

```
function [x,ni] = GaussSaidel(A,b, x0, err, it_max)
    if nargin < 5, it_max = 50; end
    if nargin < 4, err=1e-3; end
    if nargin < 3, x0=zeros(size(b)); end
    [m,n] = size(A);
    M=tril(A);
    N=M-A;
    x=x0(:);
    for i = 1:it_max
        x0=x;
        x = M\(N*x0+b);
        if norm(x-x0,inf) < err*norm(x,inf)
            ni=i;
            return
        end
    end
    error('iteration number exceeded');
```

```
end
```

```
function [x,ni] = SOR(A,b,omega, x0, err, it_max)
    if nargin < 6, it_max = 50; end
    if nargin < 5, err=1e-3; end
    if nargin < 4, x0=zeros(size(b)); end
    [m,n] = size(A);
    M=1/omega*diag(diag(A)) + tril(A,-1);
    N=M-A;
    x=x0(:);
    for i = 1:it_max
        x0=x;
        x = M\(N*x0+b);
        if norm(x-x0,inf) < err*norm(x,inf)
            ni=i;
            return
        end
    end
    error('iteration number exceeded');
end
```

Matrice e diagonal dominanta pe linii $|a_{ii}| > \sum_{j=1, j\neq i}^{n} |a_{ij}|$

Matrice e diagonal dominanta pe coloana $|a_{ii}| > \sum_{j=1, j\neq i}^{n} |a_{ji}|$

ex.:

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

Matrice diagonal dominanta => convergenta

```
function omega = relopt(A)
    M=diag(diag(A));
    N=M-A;
    T=M\N;
    if issparse(T)
        e = eigs(T);
    else
        e = eig(T);
    end
    rt = max(abs(e));
    omega = 2/(1+sqrt(1-rt^2));
end
```

PROBLEME

Problema 1

[Metoda lui Jacobi]

Problema 2

[Metoda SOR]

[Gasire omega optim]

Problema 3

```
n = 5;
A = 5*eye(n) - diag(ones(n-1,1),1) - diag(ones(n-1,1),-1)
```

```
A = 5×5
     5    -1     0     0     0
    -1     5    -1     0     0
     0    -1     5    -1     0
     0     0    -1     5    -1
     0     0     0    -1     5
```

```
b = sum(A,2)
```

```
b = 5×1
     4
     3
     3
     3
     4
```

```
[x,ni] = Jacobi(A,b,zeros(n,1),1e-8)
```

```
x = 5×1
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
ni =
19
```

```
n=1e6
```

```
n =
1000000
```

```
e=ones(n,1);
A=spdiags([-e,5*e,-e],-1:1,n,n)
```

```
A = 1000000×1000000 sparse double matrix (2999998 nonzeros)
                  (1,1)                       5
                  (2,1)                      -1
                  (1,2)                      -1
                  (2,2)                       5
                  (3,2)                      -1
                  (2,3)                      -1
                  (3,3)                       5
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     ⋮
```

```
b=[4;3*ones(n-2,1);4];
err = 1e-11;
[x,ni] = Jacobi(A,b,zeros(n,1),err)
```

```
x = 1000000×1
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
     ⋮
ni =
29
```

```
errx = norm(ones(n,1)-x)/norm(ones(n,1))
```

```
errx =
2.8823e-12
```

```
[x,ni] = GaussSaidel(A,b,zeros(n,1),err)
```

```
x = 1000000×1
    1.0000
    1.0000
    1.0000
    1.0000
```

```
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
     ⋮
     ⋮
ni =
20
```

```
errx = norm(ones(n,1)-x)/norm(ones(n,1))
```

```
errx =
9.0949e-13
```

```
wo = 1.0435;
[x,ni] = SOR(A,b,wo,zeros(n,1),err)
```

```
x = 1000000×1
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
     ⋮
     ⋮
ni =
18
```

```
errx = norm(ones(n,1)-x)/norm(ones(n,1))
```

```
errx =
5.6754e-13
```

```
A=spdiags([-e,-e,5*e,-e,-e],-3:2:3,n,n)
```

```
A = 1000000×1000000 sparse double matrix (3999992 nonzeros)
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                  (4,1)                       -1
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                  (3,2)                       -1
                  (5,2)                       -1
                  (2,3)                        5
                  (4,3)                       -1
                  (6,3)                       -1
                  (1,4)                       -1
                  (3,4)                        5
                  (5,4)                       -1
                  (7,4)                       -1
                  (2,5)                       -1
                  (4,5)                        5
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                  (3,6)                       -1
                  (5,6)                        5
                  (7,6)                       -1
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⋮
```

Problema 4

```
n=1000;
B=rand(n,5);
```

```
B(:,3) = sum(abs(B')+2);
A = spdiags(B,-2:2,n,n);
b = A*ones(n,1);
x=A\b
```

```
x = 1000×1
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
    1.0000
     ⋮
```

```
err = 1e-11;
[x,ni] = Jacobi(A,b,zeros(n,1),err);
ni
```

```
ni =
17
```

```
errx = norm(ones(n,1)-x)/norm(ones(n,1))
```

```
errx =
8.3791e-14
```

```
[x,ni] = GaussSaidel(A,b,zeros(n,1),err);
ni
```

```
ni =
12
```

```
errx = norm(ones(n,1)-x)/norm(ones(n,1))
```

```
errx =
4.1458e-14
```

```
wo = 1.0435;
[x,ni] = SOR(A,b,wo,zeros(n,1),err);
ni
```

```
ni =
14
```

```
errx = norm(ones(n,1)-x)/norm(ones(n,1))
```

```
errx =
1.2823e-13
```